

Partial Differential Egn (PDE)

one independent variable

ODEs only track how the system changes w respect to one thing at a time [mostly time]

while PDEs takes into acc >1 independent variable at a time

[Normally w respect to time (t) and spacial dimensions (x, y, z) simultaneously.]

Solution Methods

① Separable assumption [mostly work for Linear, homogeneous PDEs]

Example 1

Find solution for

$$\frac{\partial u}{\partial x} = 4 \frac{\partial u}{\partial y}$$

where $u(0, y) = 4e^{-y}$

Assume $u(x, y) = X(x) \cdot Y(y)$

$$\Rightarrow \frac{\partial u}{\partial x} = X'(x) \cdot Y(y) ; \frac{\partial u}{\partial y} = X(x) \cdot Y'(y) \quad \text{--- } \oplus$$

Sub $\oplus \rightarrow$ original ODE

$$X'Y = 4XY'$$

$$\underbrace{\frac{X'}{X}}_{\text{func. of } x} = 4 \underbrace{\frac{Y'}{Y}}_{\text{func. of } y}$$

The only way for $f(x) = f(y)$ is a constant
separation constant

$$\therefore \frac{X'}{X} = 4 \frac{Y'}{Y} = \lambda$$

$$\frac{X'}{X} = \lambda$$

$$\left(\frac{1}{X}\right) \frac{dX}{dx} = \lambda$$

$$\int \frac{1}{X} \frac{dX}{dx} dx = \lambda \int 1 dx$$

$$\int \frac{1}{X} dX = \lambda x + C_1$$

$$\ln|X| = \lambda x + C_1$$

$$X(x) = e^{\lambda x + C_1}$$

$$= Ae^{\lambda x}$$

$$4 \frac{Y'}{Y} = \lambda$$

$$4 \int \frac{1}{Y} \frac{dY}{dy} dy = \lambda \int 1 dy$$

$$4 \ln|Y| = \lambda y + C_2$$

$$Y = e^{\frac{\lambda}{4}y + C_3}$$

$$Y(y) = Be^{\frac{\lambda}{4}y}$$

$$\therefore u(x, y) = Ae^{\lambda x} \cdot Be^{\frac{\lambda}{4}y}$$

$$= Ce^{\lambda x + \frac{\lambda}{4}y} \quad \#$$

Example 2

Find particular solution for

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$$

for $0 \leq x \leq L$, $t \geq 0$

with boundary conditions $\begin{cases} u(0, t) = 0 \\ u(L, t) = 0 \end{cases}$ and initial condition $u(x, 0) = f(x)$

Assume $u(x, t) = F(x) \cdot T(t)$

$$\Rightarrow \frac{\partial u}{\partial t} = F(x) \cdot \frac{dT}{dt} ; \frac{\partial^2 u}{\partial x^2} = \frac{d^2 F}{dx^2} \cdot T(t) \quad \text{--- } \textcircled{*}$$

$\textcircled{*} \rightarrow$ Original PDE

$$F \cdot T' = (k) \cdot T \cdot F''$$

$$\frac{1}{kT} \cdot T' = \frac{1}{F} \cdot F''$$

$\therefore$ For this to be true, LHS & RHS = constant

$$\Rightarrow \frac{1}{kT} \cdot \frac{dT}{dt} = \lambda \quad \text{--- } \textcircled{1}$$

$$\Rightarrow \frac{1}{F} \cdot F'' = \lambda \quad \text{--- } \textcircled{2}$$

Solving for F , $\textcircled{2}$

When $x = 0$, $u(0, t) = 0 \Rightarrow F(0) \cdot T(t) = 0$

When $x = L$, $u(L, t) = 0 \Rightarrow F(L) \cdot T(t) = 0$

$$F(0) = 0$$

$$F(L) = 0$$

We can't set $T(t) = 0$ cuz that means no matter the time, $u(x, t) = 0$

From $\textcircled{2}$, we get $F'' - F\lambda = 0$

Auxiliary Eqn $\Rightarrow r^2 - \lambda = 0$

$$r^2 = \lambda$$

Value of discriminant	Type of roots for the auxiliary equation	Complementary function
$b^2 - 4ac > 0$	Real, distinct roots, λ_1 and λ_2	$y = Ae^{\lambda_1 x} + Be^{\lambda_2 x}$
$b^2 - 4ac = 0$	Real, repeated root, λ	$y = (A + Bx)e^{\lambda x}$ [just memorise]
$b^2 - 4ac < 0$	Complex roots: (a) $\alpha \pm \beta i$, with $\alpha \neq 0$ (b) $\pm \beta i$ (i.e. purely imaginary roots)	(a) $y = e^{\alpha x}(P \cos \beta x + Q \sin \beta x)$ (b) $y = P \cos \beta x + Q \sin \beta x$

[option 1]

Since we need to hit zero at $x=0$ & $x=L$, exponential terms is not possible because they will never have 2 x -intercepts. So we choose option 2 by making $\lambda < 0$

Let $\lambda = -(\omega^2)$ [To guarantee $\lambda < 0$]

$$\Rightarrow r = \pm \sqrt{-\omega^2}$$

$$= \pm i\omega$$

General soln is

$$F(x) = F_h + F_p$$

$$F(x) = [A \cos(\omega x) + B \sin(\omega x)] + 0$$

Beuz driving func. [RHS] = 0

Applying boundary conditions

$$F(0) = 0$$

$$A \cos(0) + B \sin(0) = 0$$

$$\boxed{A = 0}$$

$$F(L) = 0$$

$$(0) \cos(L) + B \sin(\omega L) = 0$$

$$B \sin(\omega L) = 0$$

$$B = 0 \quad \text{or} \quad \sin(\omega L) = 0$$

[Reject - Gives
trivial soln]

$$\omega L = n\pi$$

$$\boxed{\omega = \frac{n\pi}{L}}$$

$$\therefore F_n(x) = \sin\left(\frac{n\pi x}{L}\right) \quad \#$$

Solving for T, ①

$$T'_{kT} = \lambda$$

$$= -(\omega^2)$$

$$\frac{1}{T} \frac{dT}{dt} = -k \left(\frac{n\pi}{L}\right)^2$$

$$\int \frac{1}{T} dT = -k \left(\frac{n\pi}{L}\right)^2 \int 1 dt$$

$$\ln|T| = -k \left(\frac{n\pi}{L}\right)^2 t + C$$

$$T_n(t) = C_n e^{-k \left(\frac{n\pi}{L}\right)^2 t}$$

$$\therefore u(x, t) = F_n(x) \cdot T_n(t)$$

$$= \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{L}\right) e^{-k \left(\frac{n\pi}{L}\right)^2 t} \quad \#$$

How to find C_n ?

Now notice if we use the initial conditions, $t = 0$, $u(x, 0) = f(x)$

$$u(x, 0) = f(x) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{L}\right)$$

This is now a Fourier Sine Series $\left[f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L} x\right) \right]$

$$\therefore C_n = \frac{2}{L} \int_0^L f(x) \cdot \sin\left(\frac{n\pi x}{L}\right) dx$$

